

Technology Stack

Date	25 June 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit, HTML, CSS	Interactive responsive interface
2	Backend / Server-Side	Python 3.10+	Core application logic
3	Machine Learning	Scikit-learn, TF-IDF, Naive Bayes	Emotion detection
4	Generative AI	Groq API (Llama 3.1)	Personalized AI responses
5	Data Storage	CSV, Pandas	Stores interaction history
6	Data Visualization	Plotly	Emotion analytics dashboards
7	Version Control	Git & GitHub	Source code management
8	Deployment	Streamlit Community Cloud	Online deployment